ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet
HFq3RNHTGrB6yBgtwwZuLjEmGSr942PcSNUXVF46ULTD

Date: 5/1/2025
Compounds: Ibuprofen, Guaifenesin

New Compound Analysis

Name: Ibuguaifen
Formula: C13H18O3N
Weight: 250.29
Drug-likeness: 0.75

Activity:
Anti-inflammatory and expectorant

Target Analysis

Cyclooxygenase-2 (COX-2)
Binding Affinity: -8.5
Reduction of inflammation and pain

Mucin proteins
Binding Affinity: -6
Reduction of mucus viscosity

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Gastrointestinal tract

Mitigation Suggestions:
- Gastroprotective agents
- Regular liver function tests

Pharmacokinetics

Absorption:

Oral, rapid absorption

Distribution:
Widely distributed, high plasma protein binding

Metabolism:
Hepatic, via cytochrome P450 enzymes

Excretion:
Renal

Half-life: 2-4 hours

Detailed Analysis

The hypothetical compound Ibuguaifen combines the anti-inflammatory properties of Ibuprofen with the expectorant characteristics of Guaifenesin. Structurally, the molecule retains the essential pharmacophores of both parent compounds, aiming to leverage the advantageous effects on respiratory and inflammatory pathways. Mechanistically, it targets both COX enzymes, particularly COX-2, and mucin proteins to reduce mucus viscosity. This dual action is anticipated to be beneficial in conditions characterized by both inflammation and excessive mucus production, such as chronic obstructive pulmonary disease (COPD) and post-infectious cough syndromes. Predicted pharmacokinetic properties suggest rapid oral absorption and systemic distribution, which align with typical profiles of small-molecule drugs. However, moderate toxicity risks necessitate careful monitoring of hepatic and gastrointestinal health. This novel therapeutic strategy, supported by computational modeling and binding affinity data, encourages further exploration through systematic preclinical and clinical evaluations.